

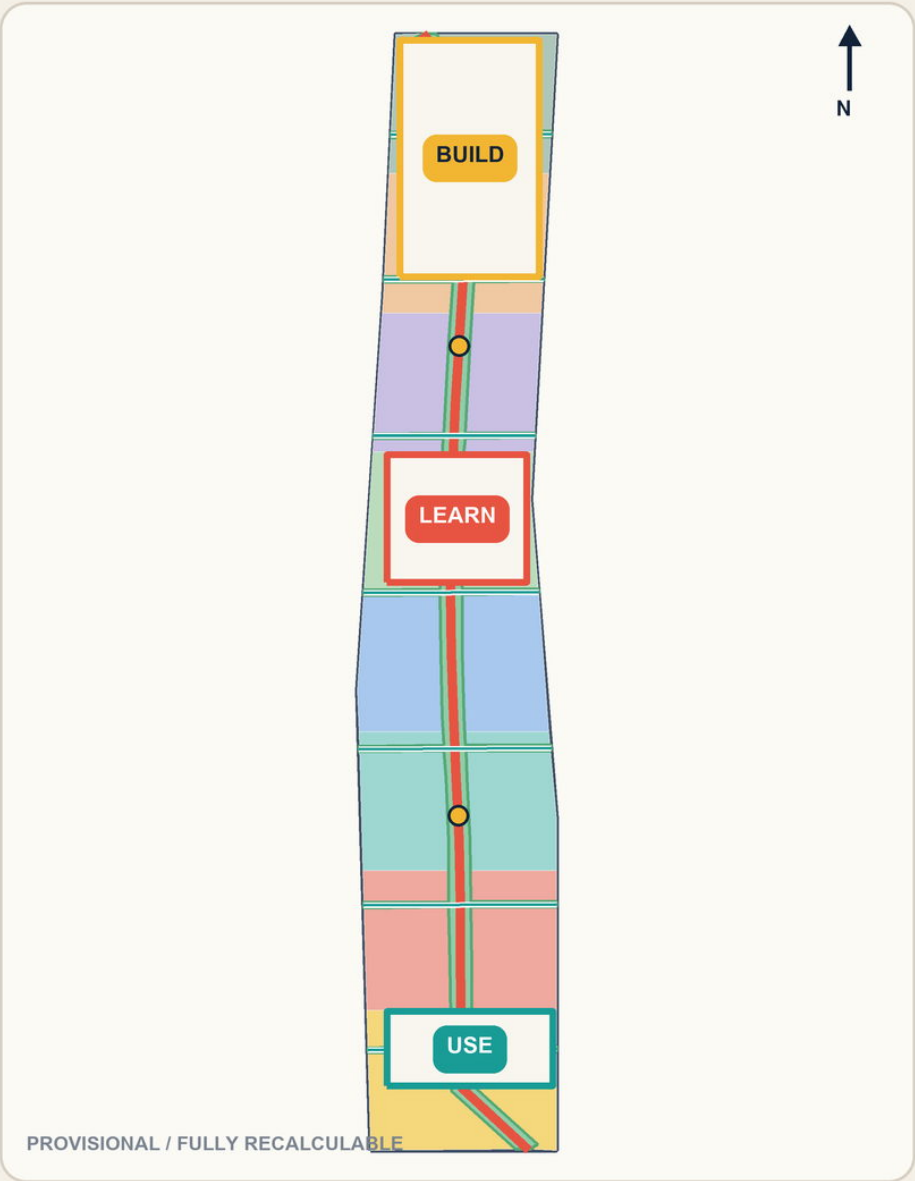
THE APPRENTICE CITY

Every AI pilot must leave the city more capable.



OPEN BOOK · RAIL · SIX GATES · RETURN

One family for the master mark, spatial signs and evidence labels—never interchangeable.



THREE POSITIONS

Centennial Jing-Zhang Cultural Belt / Metropolitan AI Life Experience Belt / AI Integrated Innovation Belt

FIVE FUNCTIONS

Full-Stack Independent AI Innovation System · World-Class AI Innovation Ecosystem · New AI+ Scenario Empowerment Paradigm · Intelligent AI-Vibrant City · Global AI Governance Voice

SIX GATES TO GRADUATION

- 1

NEED

Co-define the need, baseline and no-AI route

Need / baseline / no-AI route
- 3

PRACTISE

Simulate, contain, then practise in a limited field

Simulation → contained → limited field
- 5

TEACH-BACK

Enable a non-origin team to explain, maintain and retire it

Public lesson / two non-origin maintainers

- 2

BUILD

Declare model, data, resources, rights and limits

Model / data / rights / resources
- 4

HANDOVER

Name human roles, rehearse takeover, appeal and recovery

Takeover / incident drill / appeal
- 6

GRADUATE

A human jury decides scale, revision or retirement

Five-seat human jury: scale / revise / retire

THREE URBAN SCHOOLS

BUILD

ZHONGZHIYUAN

Controlled practice for models, compute, data, robotics and safety engineering

SC-01—04

LEARN

AI ORIGIN

Translate research into public lessons people can understand, question, learn and refuse

SC-05—08 / 10

USE

DAZHONGSI

SME adoption, operational takeover, maintenance practice, international exchange and exit

SC-09 / 13 / 15 / 16

16/16
NO-AI EQUIVALENTS · TARGET

2+
NON-ORIGIN MAINTAINERS

100
FALSIFIABLE SEMESTER / DAYS

5
HUMAN JURY SEATS



THE APPRENTICE CITY

MASTER BRAND

AI graduates only when the city can carry on.

OPEN BOOK + RAIL + SIX GATES + TEACH-BACK

THREE POSITIONS

01 CENTENNIAL JING-ZHANG CULTURAL BELT

02 METROPOLITAN AI LIFE EXPERIENCE BELT

03 AI INTEGRATED INNOVATION BELT

FIVE FUNCTIONS

F1
FULL-STACK INDEPENDENT AI
INNOVATION SYSTEM

F2
WORLD-CLASS AI INNOVATION
ECOSYSTEM

F3
NEW AI+ SCENARIO
EMPOWERMENT PARADIGM

F4
INTELLIGENT AI-VIBRANT CITY

F5
GLOBAL AI GOVERNANCE VOICE

THREE SCHOOLS: BUILD, LEARN, USE

BUILD ZHONGZHIYUAN · FULL-STACK PRACTICE YARD

Validate full-stack systems, resources and safety

LEARN AI ORIGIN · PUBLIC TRANSLATION SCHOOL

Translate research into lessons the public can question

USE DAZHONGSI · CITY ADOPTION STUDIO

Adoption, takeover, incident replay and retirement

TWO WINGS INPUT LESSONS, NOT PARTNERSHIP CLAIMS

ZHONGGUANCUN TECHNOLOGY SERVICES WING

Mentoring, legal, standards, finance and compute translation

XIAOYUE RIVER SCENARIO ENABLEMENT WING

Civic needs, climate, accessibility and operational feedback

SIX VISIBLE TASKBOOK OUTPUTS

agent.1 Brand + 3 positions + 5 functions + 3 areas/2 wings

agent.2 8 cases + 5 nodes + 8 enabler gates

agent.3 16 classrooms + 5 industry tests + 9 personas

agent.4 4 landmarks + 3 honour tracks + 7 components

agent.5 1909-2026-2126 + bilingual wayfinding

agent.6 Four seasons + maintainer chain + five-level conversion

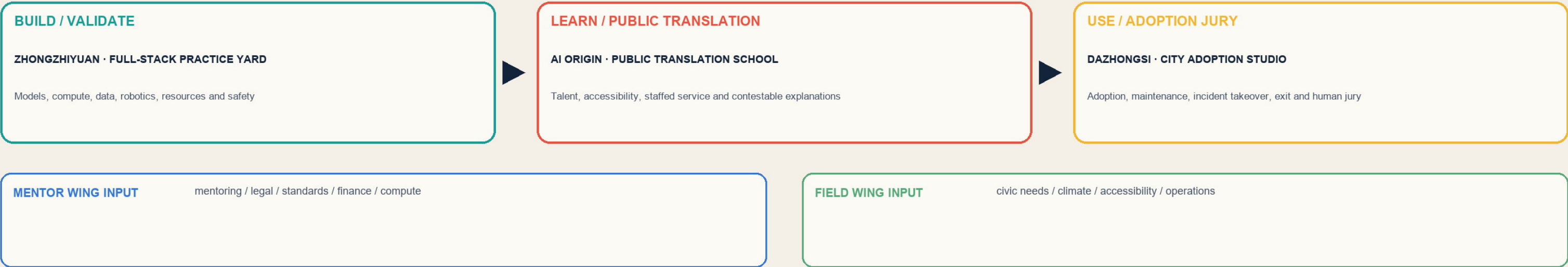
EVIDENCE: mark.svg / key_areas.geojson / scenario-cards.json / compliance_matrix.json. Every spatial expression remains a concept for professional development.

03

AI ECOSYSTEM: FIVE-NODE LOOP x EIGHT ENABLER GATES

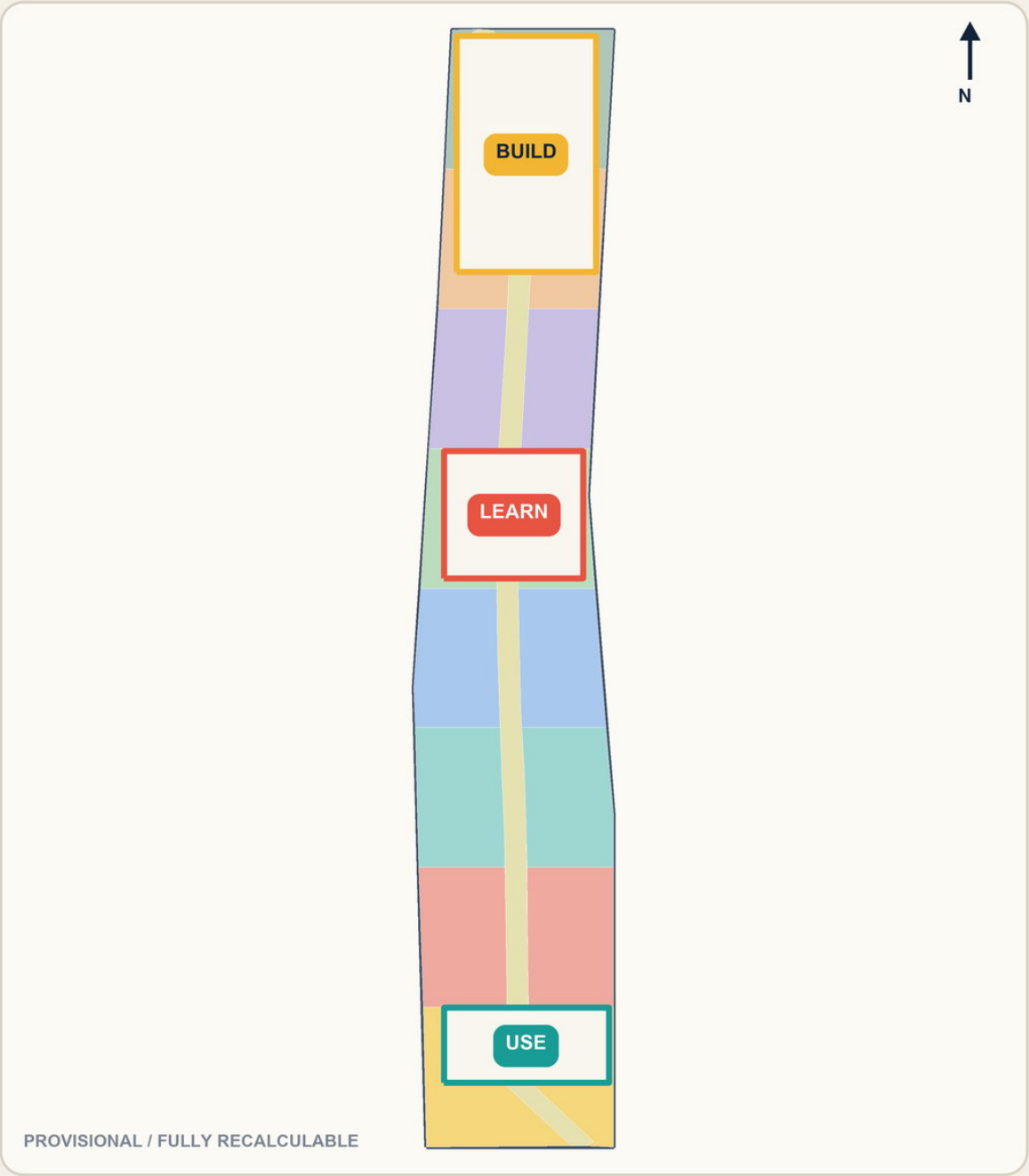
Every enabler needs evidence and a stop condition; missing capacity reduces the pilot rather than enlarging the promise.

THREE-SCHOOL LEARNING LOOP



ONE SYLLABUS · THREE SCHOOLS · TWO WINGS · SEVEN CROSSINGS

Space is a syllabus; every field practice carries a stop rule.



9
TOPOLOGY-SAFE ZONES

36
REVERSIBLE SPATIAL
PROTOTYPES

7
CROSS-CLASSROOMS

16
FIELD CLASSROOMS

THE URBAN CURRICULUM EQUATION

1
LIGHT-TOUCH
HERITAGE-PARK
LAYER

+

3
BUILD · LEARN ·
USE

+

2
MENTOR + FIELD
WINGS

+

7
EVERYDAY
CROSSINGS

+

16
GRADUATION
CONTRACTS

THE FIRST 100 DAYS / CAPABILITY, NOT CALENDAR, ADVANCES

D0—20
NEED / BASELINE

D21—45
SYNTHETIC / SIMULATE

D46—70
CONTAINED PRACTICE

D71—90
TAKEOVER /
TEACH-BACK

D91—100
HUMAN JURY

LAND USE EXPRESSES RELATIONSHIPS, NOT STATUTORY CONTROL

PUBLIC FRONT
purpose / limits / staffed route visible

CONTROLLED BACK
testing / logistics / risk contained

REVERSIBLE GROUND
ordinary use returns after AI exits

VISIBLE UNKNOWNNS
no invented FAR / height / budget

BASIC SERVICE
static / staffed / phone / paper

EXISTING BASE FIRST
retain built park / reversible learning overlay only

05 / DELIVERY & ACCOUNTABILITY

THE PILOT IS NOT THE PRODUCT; CIVIC CAPABILITY IS

Twelve actions register RACI, resources, acceptance, payment gates, degraded modes and exit.

05



P0—P3 DEPENDENCY GATES / TIME DOES NOT AUTO-ADVANCE



ACTION PACKAGES / OWNERSHIP & ACCEPTANCE

AP-01

Centennial Syllabus Park

A · Public-space steward

PASS · Reversible signs + seven field classrooms

AP-03

Public Translation School

A · Authorised service operator

PASS · Plain explanation + human service + apprentice seats

AP-05

Mentor Wing Capability Directory

A · Coordination role

PASS · Mentor/legal/standards/compute service menu

AP-07

Teach-back Graduation Bundle

A · Knowledge custodian

PASS · Schema + sample + public learning asset

AP-09

Compensated Civic Learning

A · Future operations/finance role

PASS · Compensation principle + civic seats + open selection

AP-11

Regional Train-the-Trainer Kit

A · Coordination and knowledge custodian

PASS · Lesson fields + negative results + version differences

AP-02

Full-stack Practice Yard

A · Authorised test lead

PASS · Four industry tests + public window

AP-04

City Adoption Studio

A · Authorised adoption operator

PASS · Adoption clinic + incident drill + jury

AP-06

Field Wing Classes

A · Scenario and civic liaison

PASS · Climate class + no-digital route + feedback

AP-08

Accessible and No-AI Equivalent Network

A · Service/access role

PASS · Phone + desk + paper + tactile/audio

AP-10

Four-Season Civic Semester

A · Annual programme lead

PASS · Spring needs / summer field / autumn jury / winter exit drill

AP-12

Independent Graduation Evaluation

A · Independent evaluator

PASS · Worst-group result + capability replay + exit review

GRADUATION EVIDENCE

PUBLIC LESSON

2 NON-ORIGIN MAINTAINERS

TAKEOVER DRILL

NO-AI EQUIVALENT

EXPORT / DELETE / REVOKE

RETIREMENT DESTINATION

WORST GROUP + NEGATIVE RESULT

FIVE-SEAT HUMAN JURY

CASE-AIAP

AI Singapore AI Apprenticeship Programme

TRANSFERABLE Deep-skilling followed by real projects with experienced mentors

NOT TRANSFERRED Translate learning + real work + capability replay; do not transfer eligibility, stipend, placement rate or national status

CASE-PDD

JTC Punggol Digital District

TRANSFERABLE Campus, industry, community and district platform coexist in a real precinct

NOT TRANSFERRED Use co-location and test sequence only; do not transfer land regime, sensing scale or performance

CASE-HELSINKI

Testbed Helsinki

TRANSFERABLE City, companies, researchers and residents test in real settings

NOT TRANSFERRED Translate field class and closeout report; do not copy Finnish data rights or procurement

CASE-EU-TEF

EU AI Testing and Experimentation Facilities

TRANSFERABLE Physical/virtual facilities, real conditions, end users and trustworthy-AI support form a network

NOT TRANSFERRED Translate simulation-contained-limited field gates; claim no regulatory sandbox or certification status

CASE-VECTOR

Vector Institute Applied AI Internships

TRANSFERABLE Implementation projects, mentorship and presentation connect learning to practice

NOT TRANSFERRED Use project learning and inclusive recruitment only; do not transfer eligibility, partners or outcomes

CASE-MILA

Mila

TRANSFERABLE Research, talent, entrepreneurship and responsible-AI community operate together

NOT TRANSFERRED Translate mentor and responsibility lessons; do not transfer scale, finance or governance

CASE-BCN

Barcelona Impulsa 2025-2035 / 22@

TRANSFERABLE Urban regeneration, innovation clusters, urban labs and talent appear in one agenda

NOT TRANSFERRED Use district-scale combination only; do not copy projects, law, investment or visuals

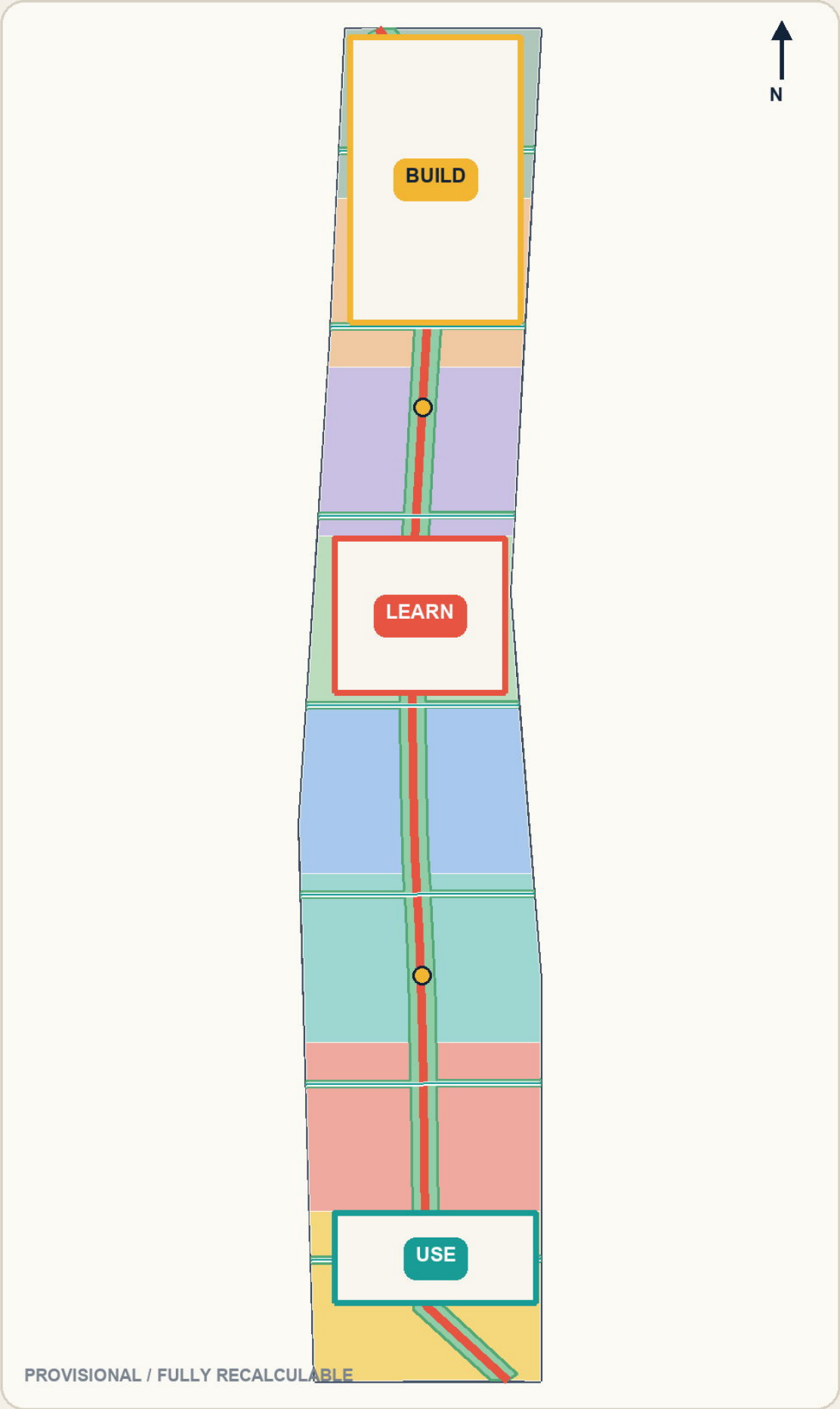
CASE-BRAINPORT

Brainport Agenda

TRANSFERABLE Triple-helix collaboration includes lifelong learning and shared prosperity

NOT TRANSFERRED Translate train-the-trainer and lifelong learning; do not transfer credential force, partners or funding

VISIBLE GAPS · official polygons / controls / buildings & ownership / road-rail redlines / utility capacity / operator / budget / real-user trials.



43.6 km² **COORDINATED RESEARCH**

who teaches whom / what capability travels

≈11.4 km² **OVERALL DESIGN**

curriculum becomes renewal, mobility, landscape and service

368.4 ha **THREE KEY AREAS**

who owns / what is practised / when to stop / how to transfer

A REGIONAL KIT CONTAINS

- 1 lesson fields
- 2 negative results
- 3 version differences
- 4 teacher guide
- 5 local revalidation gate

5 × RETEACH & REVALIDATE LOCALLY

BEIWEI COMMUNITY

Co-teach daily needs, accessibility and no-account/no-digital routes

STOP · Participation never implies deployment consent

FUTURE SCIENCE CITY

Concept exchange for research practice, talent lessons and reproducibility

STOP · No claim of existing mutual recognition

HUAIROU SCIENCE CITY

Concept link to AI for science, instruments and environmental lessons

STOP · Field use requires separate professional review

BEIJING E-TOWN

Concept link to robotics, manufacturing and engineering maintenance

STOP · No transfer of permits or industrial agreements

BEIJING-TIANJIN-HEBEI

De-located lesson kits, negative results and train-the-trainer exchange

STOP · Each place revalidates; no copying of local coordinates or approvals

02 / SPATIAL SYLLABUS

ONE SYLLABUS · THREE SCHOOLS · TWO WINGS · SEVEN CROSSINGS

Space is a syllabus; every field practice carries a stop rule.

02

BUILD

LEARN

USE

PROVISIONAL / FULLY RECALCULABLE

9
TOPOLOGY-SAFE ZONES

36
REVERSIBLE SPATIAL PROTOTYPES

7
CROSS-CLASSROOMS

16
FIELD CLASSROOMS

THE URBAN CURRICULUM EQUATION

1
LIGHT-TOUCH HERITAGE-PARK LAYER

+

3
BUILD · LEARN · USE

+

2
MENTOR · FIELD WINGS

+

7
EVERYDAY CROSSINGS

+

16
GRADUATION CONTRACTS

THE FIRST 100 DAYS / CAPABILITY, NOT CALENDAR, ADVANCES

D0—20
NEED / BASELINE

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D46—70
CONTAINED PRACTICE

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TAKEOVER / TEACH-BACK

D91—100
HUMAN JURY

LAND USE EXPRESSES RELATIONSHIPS, NOT STATUTORY CONTROL

PUBLIC FRONT
purpose / limits / staffed route visible

CONTROLLED BACK
testing / logistics / risk contained

REVERSIBLE GROUND
ordinary use returns after AI exits

VISIBLE UNKNOWNNS
no invented FAR / height / budget

BASIC SERVICE
static / staffed / phone / paper

EXISTING BASE FIRST
retain built park / reversible learning overlay only

JING-ZHANG APPRENTICE CITY · PROVISIONAL CONCEPT / NOT AN OFFICIAL REDLINE

cc-c122 · v1.0

100.00%

PROVISIONAL TOPOLOGY COVERAGE

36

CONCEPT BUILDING PROTOTYPES

UNKNOWN

FAR / HEIGHT / FLOOR AREA / BUDGET

FIVE RETAIN-ADAPT-REMOVE GATES

01

INVENTORY

ownership / age / structure / fire / heritage / use / energy

02

SAFETY & RIGHTS

geometry never substitutes field and legal evidence

03

PERFORMANCE & VALUE

continue / repair / reversible insertion

04

NEGOTIATE

affected people, operator and professionals review

05

RETAIN FIRST

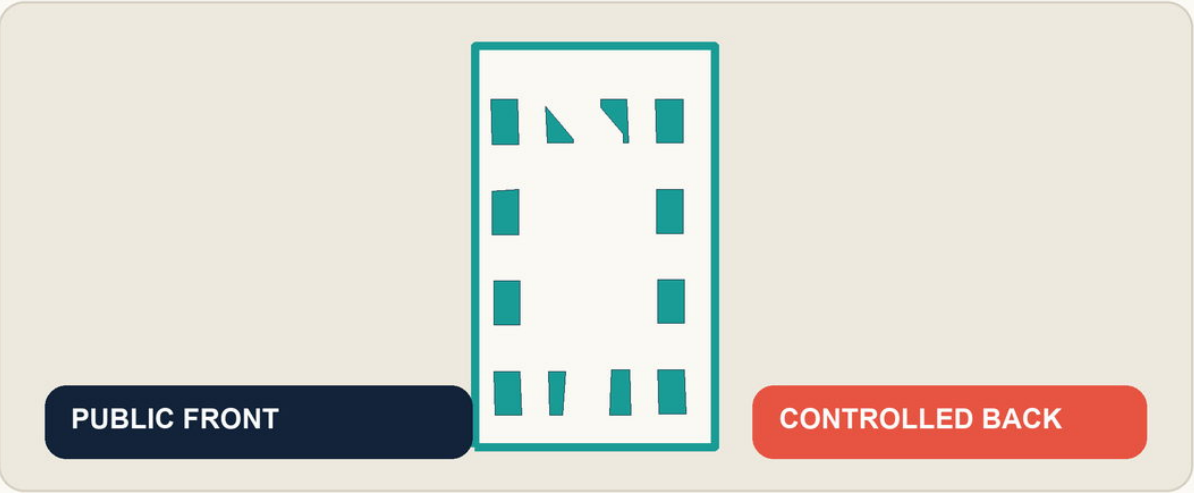
repair and adaptive reuse first; removal needs lawful proof

THE APPRENTICE CITY · cc-c122 · provisional concept

8/17

BUILD

ZHONGZHIYUAN · FULL-STACK PRACTICE YARD



SPATIAL MOVE

Shared prototype sheds, red-team rooms, resource benches, public windows and a Qinghe climate classroom

FIELD CLASSROOMS

- SC-01

Full-stack model benchmark workshop
- SC-02

Embodied-robot shift safety yard
- SC-03

Model energy and water budget class
- SC-04

Privacy-preserving data steward lab

HARD STOP

Spillover from sensitive tests, enclosure, or treating a test pass as regulatory approval

SPACE / PRACTICE / OPERATION / EXIT

PLACE

Shared prototype sheds, red-team rooms, resource benches, public windows and a Qinghe climate classroom

SC-01 Full-stack model benchmark workshop

TEACH-BACK · A non-origin team reproduces three core tests with explained variance

STOP · Unclear rights, irreproducibility or excess privilege triggers hold

SC-02 Embodied-robot shift safety yard

TEACH-BACK · An external safety operator stops, resets and removes the system

STOP · Boundary breach, failed stop, link loss or public exposure stops the test

SC-03 Model energy and water budget class

TEACH-BACK · A maintainer recomputes per-task resources and uncertainty

STOP · No metering, unknown capacity or unresolved thermal safety stops it

SC-04 Privacy-preserving data steward lab

TEACH-BACK · A second steward performs export, deletion and privilege revocation

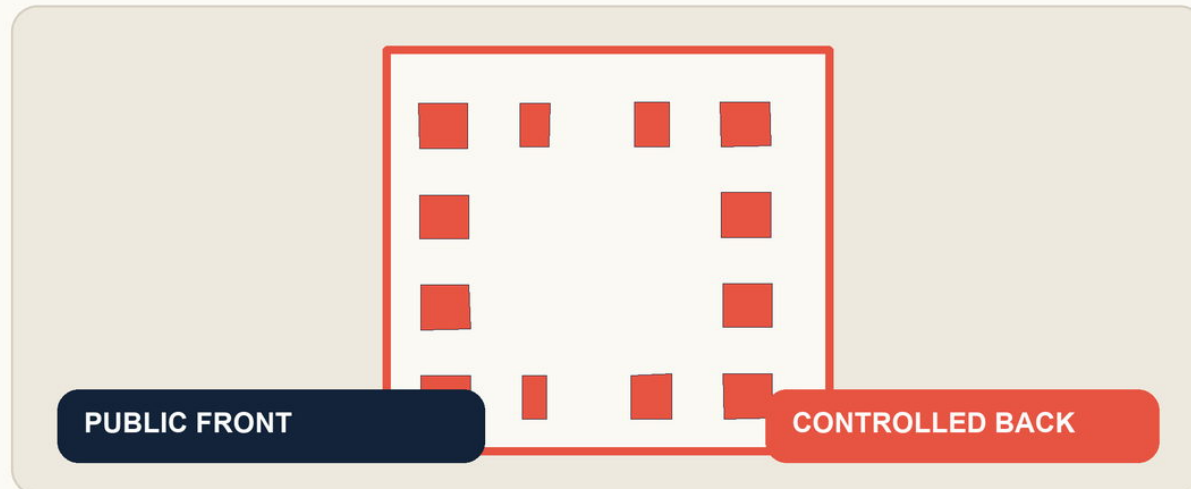
STOP · Purpose drift, re-identification risk or deletion failure triggers pause

PRIMARY RISK / PROHIBITED MISREADING

Spillover from sensitive tests, enclosure, or treating a test pass as regulatory approval

LEARN

AI ORIGIN · PUBLIC TRANSLATION SCHOOL



SPATIAL MOVE

Open seminar rooms, model translation clinics, accessibility walks, youth cars and staffed service desks

FIELD CLASSROOMS

- SC-05** AI accessibility red-team walk
- SC-06** Plain-language model translation clinic
- SC-07** Youth AI maker carriage
- SC-08** Career-shift civic project semester
- SC-10** Staffed AI public-service practice desk

HARD STOP

Default experimentation on residents, unpaid extraction, or digital substitution for basic service

SPACE / PRACTICE / OPERATION / EXIT

PLACE

Open seminar rooms, model translation clinics, accessibility walks, youth cars and staffed service desks

SC-05 AI accessibility red-team walk

TEACH-BACK · A non-origin team updates barriers, publishes detours and closes errors

STOP · Unsafe routing, access interruption or unanswered appeal stops it

SC-06 Plain-language model translation clinic

TEACH-BACK · A new operator explains purpose, limits and human route in two minutes

STOP · Unclear source, exaggerated capability or no complaint route removes it

SC-07 Youth AI maker carriage

TEACH-BACK · A peer rebuilds the work and explains data/copyright limits

STOP · Unnecessary minor data, manipulation or unsafe content stops it

SC-08 Career-shift civic project semester

TEACH-BACK · Apprentice independently performs preflight, takeover and retirement briefing

STOP · Unpaid extraction, discriminatory filtering or service overreach stops it

SC-10 Staffed AI public-service practice desk

TEACH-BACK · Backup staff complete lookup, record and escalation offline

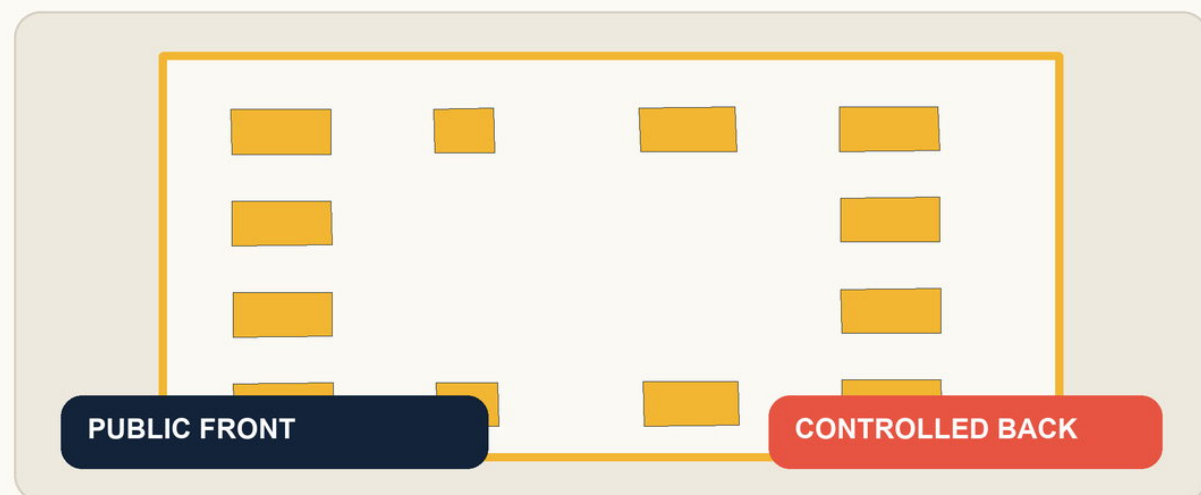
STOP · Automated high-impact decision or broken human route stops it

PRIMARY RISK / PROHIBITED MISREADING

Default experimentation on residents, unpaid extraction, or digital substitution for basic service

USE

DAZHONGSI · CITY ADOPTION STUDIO



SPATIAL MOVE

Adoption clinics, incident simulators, maintenance bays, graduation jury and station-facing services

FIELD CLASSROOMS

- SC-09** International founder city orientation
- SC-13** SME AI adoption clinic
- SC-15** City operator incident simulator
- SC-16** Adoption, maintenance and retirement jury

HARD STOP

Launch theatre, vendor lock-in, or orphaned maintenance and exit liabilities

SPACE / PRACTICE / OPERATION / EXIT

PLACE

Adoption clinics, incident simulators, maintenance bays, graduation jury and station-facing services

SC-09 International founder city orientation

TEACH-BACK · Team restates local unknowns, permit gates and non-promises

STOP · Implied incentives, deals, endorsement or unauthorised disclosure stops it

SC-13 SME AI adoption clinic

TEACH-BACK · Merchant disables the tool, exports needed records and restores manual work

STOP · Payment overreach, discriminatory pricing, lock-in or failed revocation stops it

SC-15 City operator incident simulator

TEACH-BACK · Backup shift performs stop, notice, manual service, recovery and exit

STOP · Late takeover, missing log, role conflict or failed recovery blocks graduation

SC-16 Adoption, maintenance and retirement jury

TEACH-BACK · Non-origin team explains, runs, takes over, repairs, exports and retires it

STOP · Unclear owner, sustained resource, appeal or asset exit means revise or retire

PRIMARY RISK / PROHIBITED MISREADING

Launch theatre, vendor lock-in, or orphaned maintenance and exit liabilities

P-01 Youth maker

NEED

Low-threshold practice, mentors and safe limits

NON-NEGOTIABLE

No unnecessary minor data

P-02 University researcher

NEED

Translation, reproducibility, public questioning and collaboration

NON-NEGOTIABLE

Open knowledge is not blanket release of sensitive data or IP

P-03 Startup engineer

NEED

Testing, standards, adoption and maintenance responsibility

NON-NEGOTIABLE

Payment cannot buy a pass or endorsement

P-04 Front-line operator

NEED

Configuration, takeover, recovery, staffing and dignified work

NON-NEGOTIABLE

Automation must not hide labour or accountability

P-05 Career shifter

NEED

Compensated learning, real projects and portable capability

NON-NEGOTIABLE

The civic track avoids irrelevant credential barriers

P-06 Older resident and carer

NEED

Plain explanations and phone, desk and paper equivalents

NON-NEGOTIABLE

Refusing an account or data does not reduce service

P-07 Disabled user

NEED

Continuous access, accessible formats, compensation and co-signoff

NON-NEGOTIABLE

A QR code is never the only entrance

P-08 Local SME

NEED

Affordable adoption, contract clarity, human review and exit

NON-NEGOTIABLE

No emotion pricing or opaque profiling

P-09 International visitor

NEED

Bilingual routes, legible rules and honest cooperation bounds

NON-NEGOTIABLE

Translation is appealable and cooperation is never fabricated

SC-01 Full-stack model benchmark workshop

HUMAN GATE Technical lead + independent reproducer

NO-AI Static benchmark report and staffed explanation

TEACH-BACK A non-origin team reproduces three core tests with explained variance

STOP · Unclear rights, irreproducibility or excess privilege triggers hold

SC-02 Embodied-robot shift safety yard

HUMAN GATE Two-person emergency stop and incident command

NO-AI Static exhibit and window-side explanation

TEACH-BACK An external safety operator stops, resets and removes the system

STOP · Boundary breach, failed stop, link loss or public exposure stops the test

SC-03 Model energy and water budget class

HUMAN GATE Energy/facility staff verify and may refuse expansion

NO-AI Paper label readable without running the model

TEACH-BACK A maintainer recomputes per-task resources and uncertainty

STOP · No metering, unknown capacity or unresolved thermal safety stops it

SC-04 Privacy-preserving data steward lab

HUMAN GATE Data owner role + affected-group representative

NO-AI Synthetic case route without personal data

TEACH-BACK A second steward performs export, deletion and privilege revocation

STOP · Purpose drift, re-identification risk or deletion failure triggers pause

SC-05 AI accessibility red-team walk

HUMAN GATE Disabled evaluator holds an explicit stop vote

NO-AI Tactile/static map, hotline and staff

TEACH-BACK A non-origin team updates barriers, publishes detours and closes errors

STOP · Unsafe routing, access interruption or unanswered appeal stops it

SC-06 Plain-language model translation clinic

HUMAN GATE Editor + domain owner + resident reader

NO-AI Plain paper, audio and staffed Q&A

TEACH-BACK A new operator explains purpose, limits and human route in two minutes

STOP · Unclear source, exaggerated capability or no complaint route removes it

SC-07 Youth AI maker carriage

HUMAN GATE Teacher final review with safeguarding first

NO-AI No-account kit and offline coding task

TEACH-BACK A peer rebuilds the work and explains data/copyright limits

STOP · Unnecessary minor data, manipulation or unsafe content stops it

SC-08 Career-shift civic project semester

HUMAN GATE Mentor and issue sponsor review; no automated rejection

NO-AI Open syllabus, human coaching and paper feedback

TEACH-BACK Apprentice independently performs preflight, takeover and retirement briefing

STOP · Unpaid extraction, discriminatory filtering or service overreach stops it

SC-09 International founder city orientation

HUMAN GATE Professionals decide; no automated legal/administrative conclusion

NO-AI Bilingual checklist, human booking and paper route

TEACH-BACK Team restates local unknowns, permit gates and non-promises

STOP · Implied incentives, deals, endorsement or unauthorised disclosure stops it

SC-10 Staffed AI public-service practice desk

HUMAN GATE Front-line staff own final routing and correction

NO-AI Desk, phone and paper directory always parallel

TEACH-BACK Backup staff complete lookup, record and escalation offline

STOP · Automated high-impact decision or broken human route stops it

SC-11 Low-vision and no-smartphone travel companion

HUMAN GATE On-site mobility/access staff verify safety information

NO-AI Audio button, hotline, tactile map and assisted route

TEACH-BACK Operator switches to static fallback within a 15-minute design target

STOP · Expired safety data, broken route or failed notice closes the AI layer

SC-12 Heat-and-storm adaptation street class

HUMAN GATE Professionals issue alerts; AI never replaces emergency command

NO-AI Fixed shade, static signs and human patrol

TEACH-BACK Civic steward checks sensors, sets detours and reports faults

STOP · Bad sensors, severe weather or insufficient maintenance triggers human mode

SC-13 SME AI adoption clinic

HUMAN GATE Merchant + consumer/professional role co-sign

NO-AI Manual workflow, ordinary price label and paper contract explanation

TEACH-BACK Merchant disables the tool, exports needed records and restores manual work

STOP · Payment overreach, discriminatory pricing, lock-in or failed revocation stops it

SC-14 Jing-Zhang heritage fact-check class

HUMAN GATE Archive/heritage editor decides; contributor may withdraw

NO-AI Static label, staffed tour and printed timeline

TEACH-BACK A new editor rebuilds one narrative from sources and marks uncertainty

STOP · Unknown source, unresolved rights or invented fact hides the item

SC-15 City operator incident simulator

HUMAN GATE Independent reviewer not in the original decision leads review

NO-AI Phone tree, paper manual, physical stop and assembly point

TEACH-BACK Backup shift performs stop, notice, manual service, recovery and exit

STOP · Late takeover, missing log, role conflict or failed recovery blocks graduation

SC-16 Adoption, maintenance and retirement jury

HUMAN GATE Five-seat human jury with a mandatory affected-group seat

NO-AI No-AI service and complete retirement option

TEACH-BACK Non-origin team explains, runs, takes over, repairs, exports and retires it

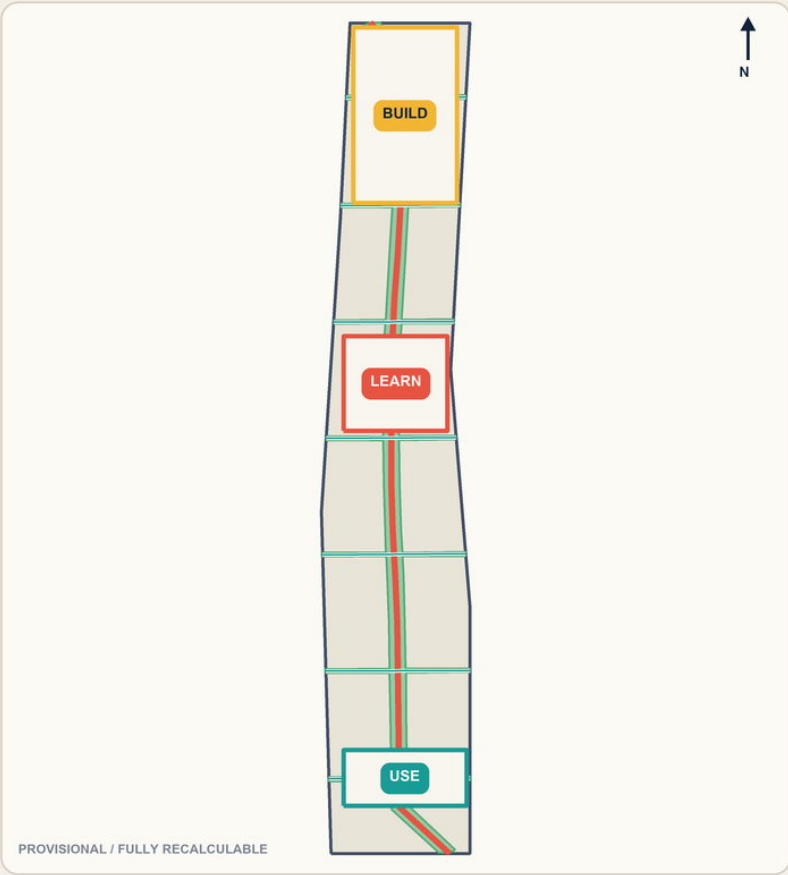
STOP · Unclear owner, sustained resource, appeal or asset exit means revise or retire

04 / PUBLIC BASE LAYER

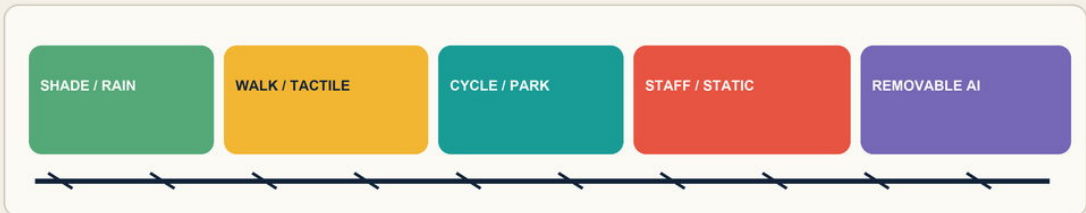
THE AI LAYER MAY GO OFFLINE; BASIC SERVICE MAY NOT

Retain the built park and fishbone walking/cycling base; add only verified crossings, staffed service and removable AI.

04



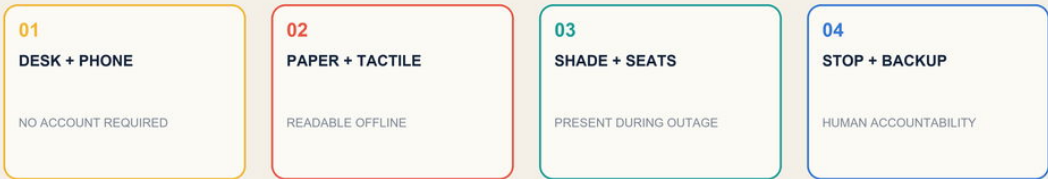
A COMMON SECTION FOR SEVEN CROSS-CLASSROOMS



THREE OPERATING LAYERS / DEGRADE WITHOUT DENYING SERVICE



FOUR FACILITIES THE AI LAYER CANNOT REPLACE



RETAIN THE BASE · Phase-two support is complete; verify seven conceptual crossings; all 16 scenarios keep no-AI routes.

JING-ZHANG APPRENTICE CITY · PROVISIONAL CONCEPT / NOT AN OFFICIAL REDLINE

cc-c122 · v1.0

THE PUBLIC BASE LAYER PRECEDES AI ENHANCEMENT

CONTINUOUS ACCESS

field walk, tactile/audio, gradients and breaks to verify

BLUE-GREEN RESILIENCE

Qinghe/Xiaoyue River interfaces, shade and stormwater need specialist study

NO-AI EQUIVALENCE

desk, phone, paper, static signs and human patrol remain

FIVE PROFESSIONAL GATES

M1 ROAD / RAIL REDLINES

statutory lines, station interfaces, sight distance, continuous access

STOP / DOWNGRADE: unknown: concept alignment only; no engineering claim

M2 POWER / COMPUTE COOLING

load curve, substation, cooling, heat rejection and equipment safety

STOP / DOWNGRADE: unknown capacity: metered bench, simulation or contained test only

M3 WATER / DRAINAGE / STORM

water capacity, flood level, outfall and sponge-asset maintenance

STOP / DOWNGRADE: missing evidence: no detention or flood-performance claim

M4 FIRE / NOISE / VIBRATION

fire access, separation, evacuation, equipment and rail noise/vibration

STOP / DOWNGRADE: failed specialist review: close, reduce or relocate the scenario

M5 MUNICIPAL INTERFACE / OPERATIONS

telecom, sanitation, charging, maintainers, incident command and shutdown

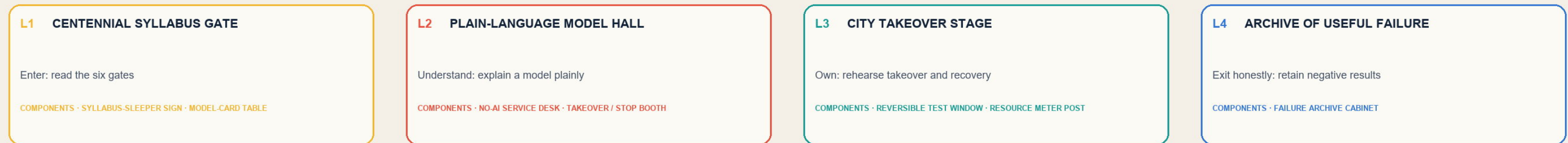
STOP / DOWNGRADE: no named operator or sustained resource: no field opening

EXISTING BASE: phase-two support is complete and opening is background only; retain the built park. Crossings, scenarios and components still need field, rights, heritage, operator and approval checks; unknown capacity keeps work simulated, contained or human-operated.

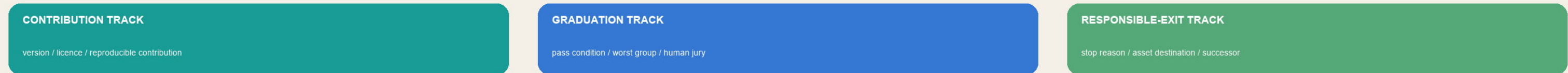
1909-2026-2126 · THREE LESSONS



FOUR CIVIC LANDMARKS / NOT MONUMENTAL OBJECTS



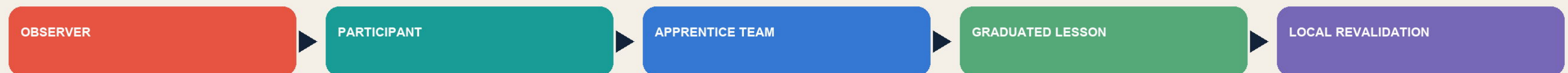
THREE EVIDENCE-BASED HONOUR TRACKS



FOUR SEASONS · PUBLISH CONTINUE, REVISE AND STOP EVERY YEAR



FIVE-LEVEL CONVERSION IS NOT A TRAFFIC FUNNEL · EVERY STEP MAY REFUSE, EXIT OR REVALIDATE



Historical facts use the National Railway Administration source only as a narrative anchor. Sites, objects, people, heritage controls, sign locations and operators still require survey, rights clearance and professional review.

FOUR OUTCOME-BASED PROCUREMENT GATES

20%

co-authored need + baseline + no-AI route

25%

synthetic/contained test + resource account

30%

two non-origin maintainers replay takeover

25%

public material + appeal + delete/export + retirement jury

TEN TRIGGER-DEGRADE-RECOVER CONTROLS

R-01

Data and privacy

TRIGGER · Purpose drift, excess fields, re-identification or failed deletion

RECOVER · Deletion/isolation proof, privilege review and independent replay

R-02

Physical and field safety

TRIGGER · Robot breach, failed stop, unsafe route or severe weather

RECOVER · Two-person stop, field check and recovery drill pass

R-03

Service exclusion

TRIGGER · No-account, older, disabled or non-Chinese user loses equivalent service

RECOVER · Worst-group field audit and affected-group co-signoff

R-04

Operations and labour

TRIGGER · No staffing, training, backup, compensation or maintenance budget

RECOVER · Named roles, labour/cost envelope and backup shift confirmed

R-05

Vendor lock-in

TRIGGER · Non-origin team cannot run, export, repair or retire

RECOVER · Six Teach-back capability replays pass

R-06

Knowledge and rights

TRIGGER · Unknown source, licence, person/mark or withdrawal right

RECOVER · Rights register, consent, citation and withdrawal complete

R-07

Planning and engineering

TRIGGER · Treat provisional geometry, concept asset or target as approved control

RECOVER · Official replacement, full-chain recalculation and professional review

R-08

Public acceptance and conflict

TRIGGER · Retaliation or no separation between initial and appeal reviewers

RECOVER · Independent decision, public de-identified finding and remedy

R-09

Technical maturity

TRIGGER · Irreproducible, excess privilege, drift or late takeover

RECOVER · Independent retest, least privilege and takeover drill

R-10

Resources and climate

TRIGGER · Unmetered energy/water/heat, unknown capacity or failed ecological maintenance

RECOVER · Resource baseline, professional capacity check and maintenance plan

This is an independent open urban-design proposal - not an official plan, permit, procurement decision, partnership or deployment approval. A valid schema proves structure, not safety or compliance.